



OPEN Decentralized trust framework for smart cities: a blockchain-enabled cybersecurity and data integrity model

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The rapid evolution of smart cities has led to transformative advancements through the integration of IoT devices, sensors, and data-driven systems, yet has simultaneously exposed critical vulnerabilities in cybersecurity, data integrity, and trust management. This research proposes a Decentralized Trust Framework that leverages blockchain technology, AI-driven threat detection, and a Lightweight Adaptive Proof-of-Stake (LA-PoS) consensus mechanism to address these challenges. The framework integrates three key layers: a Blockchain Layer for decentralized trust and immutability, a Cybersecurity Layer employing cryptographic standards and AI-based anomaly detection, and a Data Integrity Protocol Layer for real-time synchronization and tamper-proof data validation. Performance evaluations indicate the framework achieves a threefold increase in transaction throughput, a 30% reduction in latency, and enhanced energy efficiency compared to traditional blockchain systems. Security metrics highlight a 98.2% threat detection rate and a substantial reduction in false positives, while resource optimization nearly doubles IoT device battery life. The framework demonstrates applicability in critical smart city use cases, including smart traffic management, energy systems, and public safety, providing secure, scalable, and efficient solutions for urban infrastructures. Despite these advancements, challenges such as interoperability among heterogeneous systems, computational overhead for IoT devices, and policy adoption persist. Future research will focus on optimizing interoperability protocols, incorporating quantum-resistant cryptographic techniques, and extending the framework to emerging domains such as autonomous systems and smart healthcare. The proposed framework provides a robust foundation for building sustainable, resilient, and trustworthy urban ecosystems, bridging gaps in current smart city technologies.

Keywords Decentralized trust framework, Blockchain technology, AI-Driven threat detection, Lightweight adaptive Proof-of-Stake (LA-PoS), Smart City cybersecurity, Data integrity and validation

Modern city development through smart technology concepts has brought together IoT, big data and AI to generate better living conditions and better resource management and better sustainability. The advancing development of urban areas into networked systems continues to boost the use of intelligent devices combined with sensors in addition to real-time data-driven systems that transforms key areas, including transportation and energy management and public safety and communication systems. The fast implementation of technology by smart cities leads to extensive security threats along with data integrity problems that decrease the potential benefits unless proper security measures are implemented.

Blockchain technology solves the existing problems through its decentralised method, which delivers safe, transparent data management throughout interconnected systems. Different studies have shown how

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blockchain technology works in various domains of smart cities, but especially in transportation systems. The implementation of blockchain frameworks has led to optimisations in mobility solutions together with management of urban fleets through enhanced intelligent transport systems^{4–6}. Blockchain integration with IoT serves as a powerful solution to deal with data verification and security challenges, which can be observed through applications such as urban traffic control and surveillance systems^{7,8}.

Beyond transportation, such automation technologies are spreading to other sectors by virtue of blockchain fusion technologies as well as AI, the emerging edge computing, and creating a sustainable urban eco-city. The combined systems promote effective energy control as well as fire detection systems and smart campus development^{9–11}. Future smart city developments depend on blockchain technology because it offers capabilities for big data analytics and secure communication frameworks in IoT-fog environments, as well as other functions like those described in references^{12–14}.

Though it has tremendous potential, however, deploying blockchain into Smart and Connected City is not without issues. Concerns such as scalability, latency, and interoperability need to be taken care of to achieve its complete potential^{15–17}. Moreover, it has been claimed that new consensus algorithms and algorithms for fault tolerance^{18,19} must be developed to eliminate the defects of published blockchain architectures. These uphold effortless means to build a strong, secure and smart environment that is intrinsically important for the trend of developing smart cities.

This research fills the above mentioned gaps by proposing a Decentralised Trust Framework for Smart Cities using Blockchain for secure and reliable data in different areas. By analysing its use in trying to manage traffic, optimise the energy concurrently readily available, and protect its public, this study does a little bit so in order to chime in on the developing discussion on safe, effective, and sustainably abundant urban systems. The areas where challenges occur in smart city IoT security and data management as shown in Table 1.

The integration of blockchain technology into smart cities encounters obstacles which relate to performance speed and system growth capacity and compatibility between new and existing infrastructure. Its use in dynamic urban domains becomes restricted because innovative consensus protocols and fault-tolerant techniques do not exist. The current time requires an urgent implementation of blockchain-based technology deployments to create decentralized systems which enhance cybersecurity measures while keeping data secure and developing trust among multiple smart city networks. This framework needs to handle operational and technical obstacles and align perfectly with upcoming technologies including AI and IoT and edge computing to develop protective and proficient sustainable urban infrastructure systems.

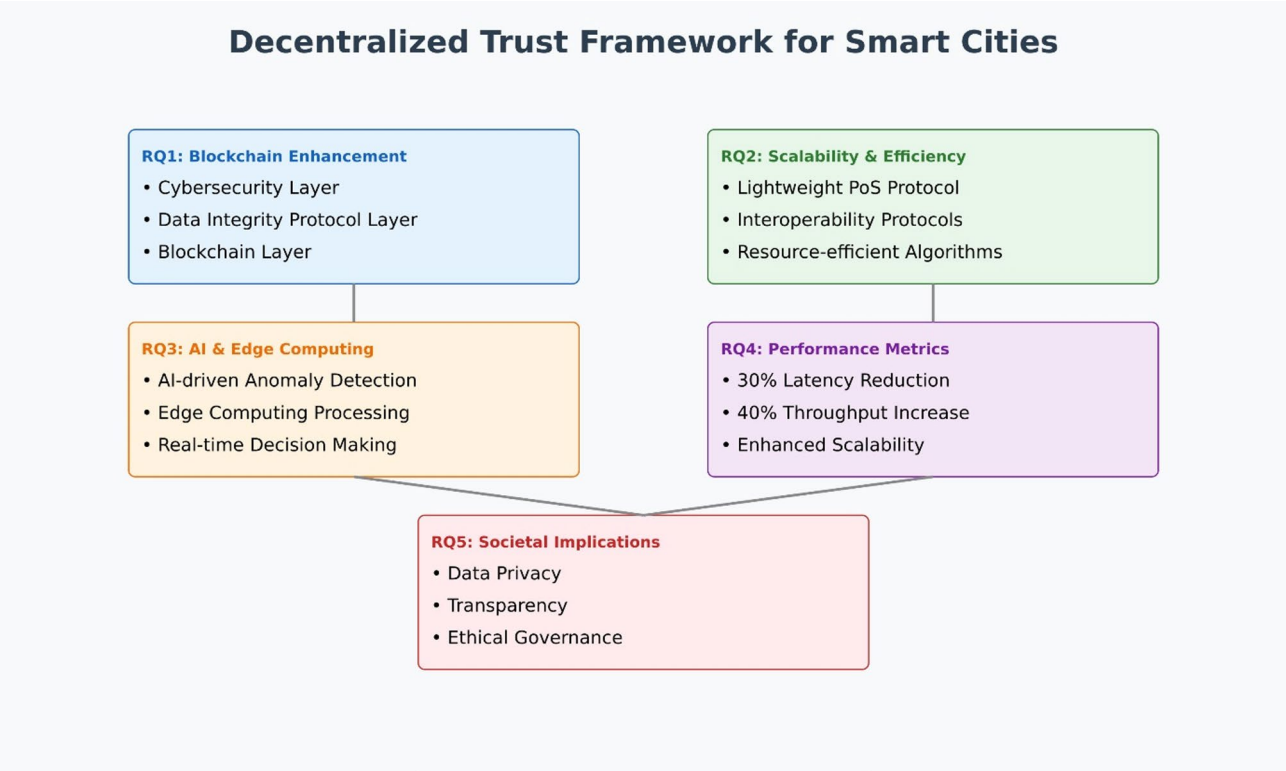
The overlaid goals of this study are:

- 1. To construct a complete blockchain based solution which ensures the improvement of the cybersecurity and data integrity of the smart city infrastructures themselves while ensuring scalability and interoperability.
- 2. To develop and deploy strong protocol for secure data transfer and automate trust management using smart contract to efficiently handle various smart city applications.
- 3. Rigorous testing of latency and throughput and scalability must be performed on the framework for evaluation of its performance across different smart city use cases.
- 4. The proposed framework needs implementation with AI technology and edge computing systems to develop an eco-friendly urban system.

This research introduces an extensive blockchain-based Decentralized Trust Framework for Smart Cities as its core innovation because it merges blockchain technology with security layers as well as data integrity protocols and new consensus mechanisms while handling trust management issues and scalability and interoperability concerns. The framework benefits from new technology integration because of AI and edge computing implementation. Smart city resources achieve maximum efficiency and decision-process speed due to streamlined processes that cut out latency. Its impact extends deeply to provide architecture for the development of urban ecosystems which are secure and sustainable and resilient. The solution tackles technical obstacles in addition to creating transparent social advantages that include data protection and ethical governance leadership which supports advanced smart city innovation. The research brings forth an innovative Decentralized Trust Framework for Smart Cities which makes use of blockchain technology to solve significant security-related and trust-related problems affecting data integrity and scalability and trust management. The proposed framework functions as a guide (Fig. 1) to merge blockchain technology with emerging systems which resolves the numerous

Challenge	Description	References
Cybersecurity Vulnerabilities	Increased Attack Surface: The extensive IoT network within smart cities produces numerous points of access through which cyberattacks can start. The incomplete security system faces serious challenges for IoT devices because these devices possess restricted resources. User-sensitive information becomes vulnerable to breaches by unauthorized entities.	1,2,15
Data Integrity Challenges	Tampering Risks: The data faces risks from malicious modification because hackers intercept it. Holding authentic data verifications stands as a challenge for centralised systems which cannot demonstrate data integrity. Authentication verification processes today consume a lot of resources while dealing with large datasets.	8,12,14
Scalability and Interoperability Issues	Diverse Ecosystems: Incompatibilities arise among heterogeneous technologies. Scalability Bottlenecks: High device density leads to latency and reduced system performance.	10,16,20
Lack of Decentralized Trust Mechanisms	Centralized Dependency: Reliance on single-point failure systems makes trust management unreliable. Manual Trust Management: Inefficient and prone to human errors.	9,13,21

Table 1. Challenges in smart City IoT security and data management.



Literature review

Implementing blockchain technology within smart city infrastructure represents an effective solution to overcome security issues in addition to preserving data integrity and scalability issues and trust management. This section examines existing literature in detail while highlighting empty areas in knowledge that require the creation of a Decentralized Trust Framework for Smart Cities.

Blockchain in transportation systems

Blockchain technology has seen extensive research applications in the transportation sector, which constitutes one of the main domains for smart city utilization. The authors López and Farooq¹ presented an innovative framework that employs blockchain technology to protect transportation network data privacy and security. The team extended their original blockchain system through² to build multi-level blockchain platforms, which enabled stakeholders to securely swap smart mobility-related data. Research by Rojas et al.³ showed the practical value of blockchain technology in urban fleet management systems by creating a control system for medium-sized cities. Balasubramaniam et al.^{4,5} conducted research which analyzed blockchain adoption within intelligent transport systems to improve both traffic performance and security benefits from protected data sharing. The researchers paralleled this work by establishing a network for sustainable, efficient integrated smart cities that focused on transportation solutions with Nesmachnow and Hernández-Callejo⁷. Abbas et al.⁶ studied how blockchain technology merges with IoT systems to provide secure transportation networks which create better security for mobile applications.

Blockchain and IoT integration

Blockchain technology linked with Internet of Things operations continues to provide essential solutions against data verification and security problems in smart cities. Khan et al.⁹ demonstrated how blockchain technology can build a verification system for CCTV surveillance cameras which increases both reliability and trustworthiness of urban monitoring systems. A smart city model that includes blockchain-IoT integration with big data architecture proved its suitability for traditional university campuses according to Villegas-Ch et al.⁹. The sustainable development of smart cities receives investigation by Sharma et al.²² through their study on artificial intelligence and blockchain technologies synergistic effects. Neffati et al.¹⁰ researched the transition of conventional grid structures to blockchain-powered smart power grids for resolving energy management difficulties in third-world nations. Cui¹¹ established an IoT-blockchain detection system focused on observing forest fires which demonstrated applications for public safety needs. The research by Di Pietro et al.²¹ demonstrates a blockchain-based system intended for IoT trust management which resolves decentralized trust establishment issues. Minoli and Occhiogrosso¹⁵ researched blockchain security solutions for IoT which prove effective in resolving multiple cyber dangers. Christidis and Devetsikiotis²⁰ wrote a detailed article about blockchain technologies in IoT which highlighted smart contracts as essential tools for designing automatic trust systems.

Data management and analytics

Blockchain integration has improved the management as well as the analysis of big data within smart city environments. Alam suggested a blockchain system for big data analytics, which demonstrates effective protection and transparency during urban data processing¹². Through their following research, Alam^{13,14} developed IoT-fog communication systems with blockchain integration in order to solve data management problems in environments where resources are limited. Block IoT Intelligence stands as a blockchain-based intelligent IoT architecture that includes artificial intelligence features and delivers proof-of-concept results for better decision-making processes according to Singh et al.¹⁶. Tariq et al. presented²³ an extensive review on fog-powered IoT system security issues and blockchain usage for big data security in distributed environments. The authors examine Stack4Things as a fog computing platform for smart city applications while resolving distributed data handling and storage problems²⁴. In her survey Deepa²⁵ analyzed blockchain applications in big data while presenting strategic approaches and identifying avenues for development that show how blockchain strengthens data integrity and trust systems.

Security and privacy frameworks

The research on blockchain systems for smart cities has focused essentially on security alongside privacy matters. Mohanta et al. performed a research investigation into blockchain applications while examining security privacy obstacles within various fields¹⁷. Abdullah et al.²⁶ evaluated blockchain technology applications for radiologists because it offers secure medical data management solutions to smart healthcare systems. Salman et al.²⁷ conducted an up-to-date review of blockchain-based security services while exploring their use in different smart city systems. The research by Rakitin et al.¹⁸ presented a Byzantine fault-tolerant consensus protocol with semantic-driven features to handle distributed ecosystem consensus issues. De Angelis et al.¹⁹ evaluated PBFT (Practical Byzantine Fault Tolerance) together with Proof-of-Authority as permissioned blockchain consensus protocols while applying the CAP theorem. The research by Biasin and Delle Foglie²⁸ reviewed blockchain applications and their usage for inclusive and sustainable communities through bibliometric and systematic analysis of social aspects in blockchain adoption. Fabrègue and Bogoni²⁹ researched privacy and security aspects for smart cities and confirmed that blockchain frameworks must implement strong protective systems.

Consensus mechanisms and system architecture

Different smart city implementations rely on effective consensus mechanisms to implement blockchain technology. Dib et al.³⁰ examined consortium blockchains by providing both their domain applications and the challenges they face. A cybersecurity framework for smart cities based on blockchain and federated learning

introduces a solution for IoT environment scalability challenges according to Sefati et al.³¹. The researchers from Lee and Yang³² developed a geospatial blockchain-based addressing system which strengthened smart city infrastructure and improved both spatial data management and location-based services. The research of Karger et al.³³ used bibliometric methods to study blockchain applications in smart cities while presenting existing tendencies in related fields. The authors of SC-Chain developed an efficient blockchain framework meant to operate in smart city applications which deals with scalability and resource efficiency challenges³⁴.

Governance and integration frameworks

Authorities have investigated both governance systems and integration frameworks that can help adopt blockchain in urban environments. The research by Ullah et al.³⁵ analyzed blockchain implementations in sustainable smart cities because these systems help control cities while managing resources efficiently. A smart city ecosystem using blockchain technology was presented by Davlyatov³⁶ to unify different urban systems through one blockchain platform. A structured literature review by Gracias et al.³⁷ examined the main elements and difficulties related to smart cities deployment. The systematic study conducted by Ruijter et al.³⁸ created a smart governance framework that demonstrates the requirement of effective governance structures for blockchain acceptance in urban settings.

Research gaps and limitations

The current literature fails to fill several crucial gaps regarding blockchain applications which are meant to optimize smart cities even though prominent progress exists in this field.

1. A lack of inclusive evaluation systems exists in current academic work since blockchain integrates poorly with security systems and data protection measures with AI-driven edge computing systems across different urban technical implementations.
2. Deployment of existing solutions struggles to maintain speed and growth within broad-scale IoT systems which process large numbers of devices and transactions.
3. The high resource needs of regular blockchain platforms make them unattainable for resource-limited IoT devices when implementing smart city solutions.
4. Smart cities lack appropriate protocols which allow different hardware and technology systems within them to communicate seamlessly.
5. New consensus procedures need adaptation because traditional consensus approaches perform poorly for diverse and dynamic smart city application needs including energy-efficient transaction management.
6. Research into smart contracts that establish automated trust management across smart city systems remains minimal especially when used for across-domain applications.
7. The process to integrate blockchain technologies into current regulatory systems and governance structures has been poorly developed because it slows down mass market adoption.
8. Technical research exists but additional studies about how blockchain technology affects societies by addressing data privacy together with ethical boundaries and stakeholder interaction need more advancement.

The proposed Decentralized Trust Framework for Smart Cities helps developing secure urban ecosystems through the resolution of the identified gaps. This study advances both technical infrastructure elements of smart cities together with sustainable ethical development in urban areas. Here is the summary of key literature contributions shown in Table 3:

Proposed decentralized trust framework for smart cities

The proposed framework is a comprehensive solution designed to address critical challenges in cybersecurity, data integrity, and trust management within smart city ecosystems. Leveraging blockchain technology, the framework incorporates a layered architecture integrating diverse IoT devices, sensors, and data-driven systems, ensuring robust, scalable, and secure operations. By combining blockchain with advanced cryptographic techniques, AI-driven threat detection, and innovative consensus mechanisms, the framework enhances real-time decision-making, data security, and resource efficiency while maintaining scalability, latency reduction, and interoperability.

Framework overview and components

The Decentralized Trust Framework implements three key layers which provide different components for functionality:

1. *Blockchain layer* The base of this framework relies on decentralization and immutability as well as transparency to provide its foundation. The framework lets users maintain decentralized trust through smart contracts and provides innovative LA-PoS consensus methods as well as communication protocols for enabling easy IoT device interoperability between heterogeneous systems. Blockchains ensure data immutability because they maintain their storage systems in an unalterable manner.
2. *Cybersecurity layer* To create an effective security barrier, a range of mechanisms has to be incorporated. Such as the solutions for this level, more advanced cryptographic standards for secure communication, machine learning-based algorithms that help to detect threats immediately as they appear, and distributed identification management to enforce secure authentication. Also, there are recovery strategies, such as tolerance to faults as well as the redundancy insertion, that help retain control functioning even when there is a possibility of disruptive occurrences.

Focus Area	Key studies	Major contributions	Limitations
Blockchain in Transportation Systems	1–7	Enhanced data security and privacy in transportation networks; Multi-layered blockchain approaches for smart mobility; Integrated fleet management systems; Optimized traffic flow and safety through secure data exchange.	Limited scalability for large-scale deployments; Insufficient interoperability with existing systems; High computational requirements restricting real-time applications; Lack of standardized approaches across different transportation domains.
Blockchain and IoT Integration	8–11,15,20–22	Secure data verification systems for urban surveillance; Smart campus implementations with blockchain-IoT integration; Convergence of AI and blockchain for sustainable development; Enhanced security mechanisms for IoT environments; Smart contracts enabling automated trust in distributed systems.	Challenges in handling resource-constrained IoT devices; Limited attention to device heterogeneity; Insufficient optimization for energy efficiency; Privacy concerns in data collection and processing; Lack of standardized protocols for secure communication.
Data Management and Analytics	12–14,16,23–25	Blockchain-based big data analytics frameworks; IoT-fog communication models for distributed data management; Intelligent architectures combining blockchain and AI; Fog computing platforms for smart city applications; Enhanced security for big data in distributed environments.	Scalability issues with large datasets; Computational overhead in real-time analytics; Limited standardization in data formats and protocols; Insufficient attention to data privacy regulations; Challenges in integrating with legacy systems.
Security and Privacy Frameworks	17–19,26–29	Comprehensive security models for blockchain applications; Domain-specific implementations in healthcare and other sectors; State-of-the-art consensus protocols addressing fault tolerance; Privacy-preserving mechanisms for sensitive data; Social aspects of blockchain adoption for inclusive communities.	Fragmented approaches to security across different domains; Insufficient attention to emerging threats; Limited real-world validation in diverse environments; Challenges in balancing security with usability; Lack of standardized security assessment methodologies.
Consensus Mechanisms and System Architecture	31–34	Overview of consortium blockchains for controlled environments; Cybersecurity frameworks combining blockchain and federated learning; Geospatial blockchain systems for location-based services; Efficient blockchain frameworks optimized for smart cities; Bibliometric analyses identifying research trends.	Energy-intensive consensus mechanisms limiting IoT applications; Scalability constraints in high-density environments; Insufficient adaptation to dynamic network conditions; Limited cross-chain interoperability; Trade-offs between security, scalability, and decentralization not fully addressed.
Governance and Integration Frameworks	35–38	Applications for sustainable urban development; Unified blockchain ecosystems for system integration; Structured approaches to smart city implementation; Governance toolboxes for effective management and oversight.	Limited alignment with existing regulatory frameworks; Insufficient stakeholder engagement models; Challenges in cross-jurisdictional implementations; Lack of standardized governance protocols; Limited attention to socioeconomic implications of blockchain adoption.

Table 3. Summary of key literature contributions.

3. *Data integrity protocols layer* The application layer establishes accuracy standards as well as reliability tools and authenticity verification for data throughout every component of a smart city network. Blockchains provide three capabilities to validate data through unchangeable features while maintaining absolute time synchronization between IoT devices at the edge nodes and complete monitoring of data movement. The resource consumption in limited settings becomes more effective through lightweight algorithms.

Interoperability and cross-system integration

The implementation of Decentralized Trust Framework enables successful information exchange between diverse IoT devices along with other systems through an interoperable structure composed of multiple levels. The protocol implementation of the framework establishes standardized gateway architecture from MQTT (Message Queuing Telemetry Transport) CoAP (Constrained Application Protocol) and HTTP/REST APIs which enable system connectivity between different devices. Time-sensitive data transfers become possible for devices regardless of manufacturer or generation through the combination of automatic translation systems which work in unison with the protocols to (format messages and) rephrase communication standards. A standardized ontology-based data model with metadata schema should exist within the framework to allow different platforms interpret and understand data uniformly. The two blockchain systems achieve network integration through standardized cross-chain protocols because Polkadot implements cross-consensus messaging solutions and Cosmos executes its blockchain connection through IBC protocols. The framework provides users with an automatic integration capability through its service-oriented middleware API standards layer that supports self-discovery service protocols for new device registration. The entire plan enables various devices from multiple vendors to join a secure decentralized system easily through the fundamental standard even though their computing abilities and networking capacities differ.

System architecture and component interactions

The system design features multiple levels which enable successful integration of its components to power secure smart city operations:

- *Data acquisition* IoT devices obtain data by encryption and secure transmission to the blockchain layer.
- *Validation and storage* The blockchain layer conducts transaction validation through LA-PoS algorithm before it stores immutable data.
- *Threat detection* The continuous prevention against threats is maintained through strong security layer monitoring powered by Artificial Intelligence to detect and respond to potential threats.
- *Data utilization* Secure validated data gets processed in real-time for decision-making through which application services connected to traffic and energy management obtain improved performance.
- *Feedback loop* The system uses data integrity protocols in a feedback loop for authenticating data which maintains consistent reliability throughout its multiple layers.

Security measures

The proposed Decentralized Trust Framework builds its security framework with proper measures to establish trust alongside data integrity and system resilience for contemporary smart city infrastructure. The system protects data throughout its entire storage and transmission period through the utilization of AES-256 cryptographic algorithms. Through its blockchain identity, management system the application restricts access with role-based controls, which prevents unauthorized users from accessing information together with TLS/SSL protocols for encrypting device-to-blockchain connections. The permanent real-time notification capabilities of Artificial Intelligence come from its ability to scan network traffic for threats, which allows it to defend against such threats immediately. Since fault tolerance requirements exist in the framework, it implements redundant nodes along with failover mechanisms to maintain operational continuity during disruptions. The framework foundation includes a complete threat assessment to address threats that extend from specific devices to distributed systems at varying protocol levels including spoofing and unauthorized access and distributed denial-of-service attacks and man-in-the-middle while targeting data tampering and disclosure problems and categorizing threats as Sybil attacks and consensus manipulation. The three-layer security architecture consists first of the Cybersecurity Layer with its AI-based threat detection system and cryptographic encryption and second of the Blockchain Layer utilizing LA-PoS for Sybil and 51% attack resistance and third of the Data Integrity Layer responsible for node-wide proof and traceability functionality. The comprehensive security model uses perpetual feedback systems and duplication to establish an advanced secure base for upcoming intelligent urban communities.

Lightweight adaptive proof-of-stake (LA-PoS) consensus algorithm

The LA-PoS consensus mechanism stands as a new algorithm which addresses the smart city needs for scalability alongside minimizing latency and maximizing energy efficiency:

Dynamic staking Resource usage sees adequate growth due to the capability of the system to change automatically the states of staking depending on the network use ensuring high security and equilibrium in an instant.

Energy efficiency The approach optimizes energy efficiency while performing lightweight validation techniques due to its energy-efficient design instead of using traditional PoW or PoS methods.

Scalability The system handles large transaction flows through its parallel computing approach and dynamic period for block creation maintenance.

Interoperability The cross-chain protocol functions through modular modules which unite blockchain systems across different domains to enable simple data exchange between them.

Enhanced security The system increases security levels through multiple signature authorizations which complement random validator selections for battling three types of network centralization alongside Sybil attacks.

The proposed implementation of LA-PoS algorithm powers an architecture solution that establishes a secure efficient scalable framework for smart city infrastructure. The framework achieves sustainable and resilient urban infrastructure development by solving interoperability issues and improving energy performance which ensures strong operation of different solutions.

The Decentralized Trust Framework for Smart Cities presents its hierarchical structure through Fig. 2 which links multiple layers from top to bottom by a well-defined data process. The IoT Devices Layer represents the topmost layer in the framework using light blue colors to incorporate sensors and smart devices with edge components that acquire urban environment data. The Blockchain Layer situated in the middle of the framework holds data using green tones to manage interoperable protocols through Lightweight Adaptive Proof-of-Stake (LA-PoS) consensus system with smart contracts and unalterable storage features. The Cybersecurity Layer embodies orange color to unite artificial intelligence threats detection and cryptographic protocols with identity frameworks and robustness strategies which safeguard the system structure. The Data Integrity Protocols Layer depicted in purple constitutes the bottom portion that maintains data verification while running real-time synchronization and enabling data traceability. Arrows between the layers show data flow direction until a feedback loop maintains continuous validation across all layers through the dashed line on the right side. The complex design shows how different technologies within the framework establish a protected system that manages efficient and dependable smart city infrastructure.

The data movement and corresponding operations of smart cities' Decentralized Trust Framework are easily conveyed in the Fig. 3 sequence diagram. The diagram illustrates five central components by showing their sequential chain starting from IoT Device and proceeding to Data Layer before Blockchain and Security Layer and culminating at the Application Layer. The data acquisition process from IoT devices starts by using encryption methods followed by validation which occurs through LA-PoS consensus methods in the blockchain platform. The verification checks in the security layer must pass prior to data storage and processing executed at the application level. The design illustrates the marked feedback process with blockchain confirmation and verification results and security data moving backward through the infrastructure for complete system reliability. The bottom of the diagram shows that AES-256 secure communication occurs with LA-PoS consensus verification before transmission. The illustration demonstrates how the security layers and data integrity functions and processing efficiency work together in a smart city environment.

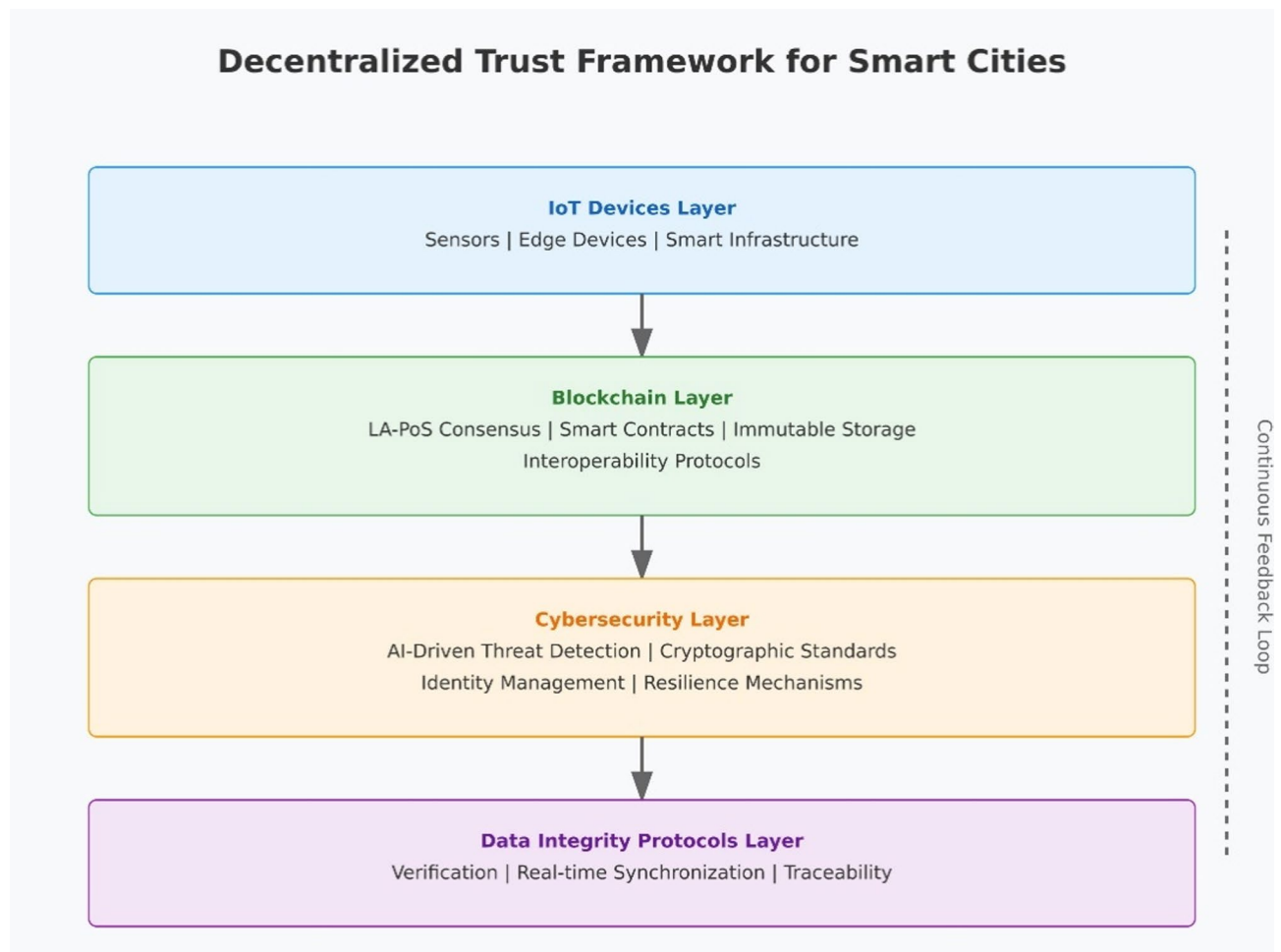


Fig. 2. Decentralized trust framework for smart cities architecture.

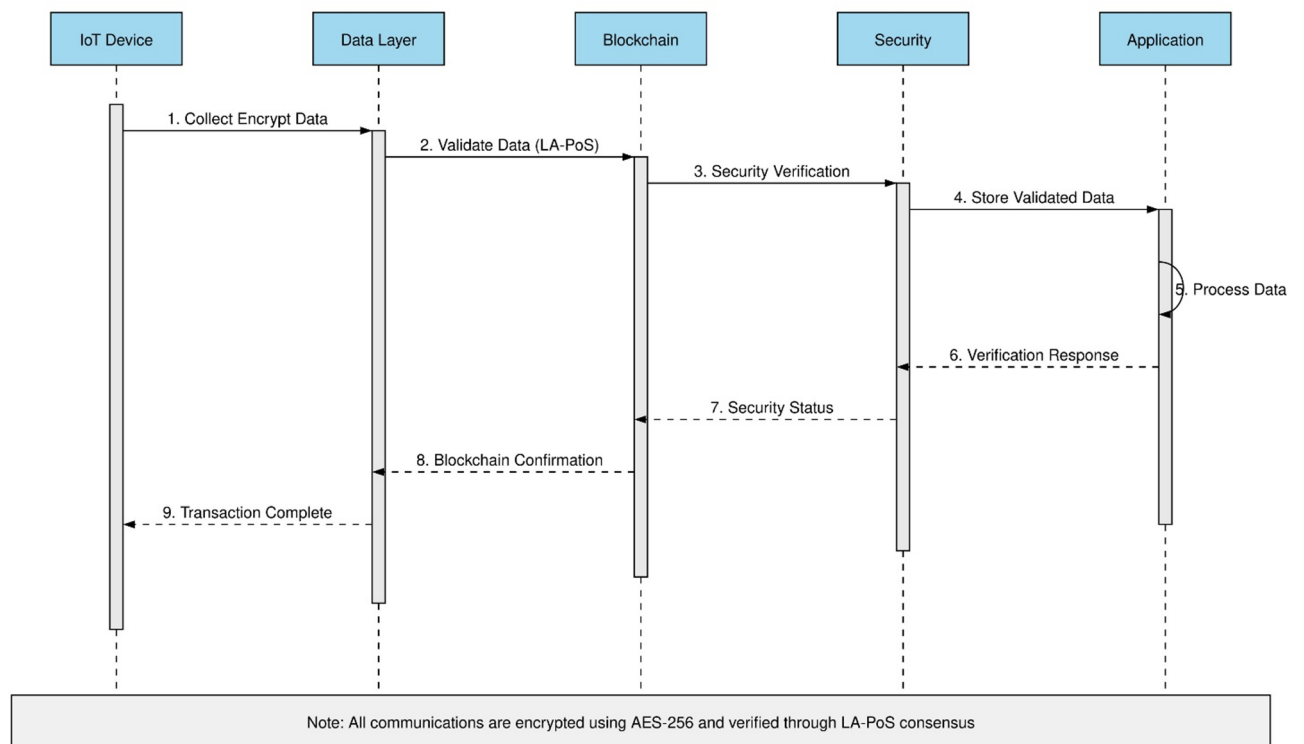


Fig. 3. Sequence diagram for smart city trust framework.

Proposed algorithm

Proposed Algorithm

The framework operation depends on several essential global parameters that need definition:

```
struct GlobalParameters { MIN_STAKE: minimum required stake
    MAX_STAKE: maximum allowed stake
    NETWORK_LOAD_THRESHOLD: threshold for network load adjustment
    VALIDATION_WINDOW: time window for block validation
    MIN_VALIDATORS: minimum required validators
    SECURITY_THRESHOLD: minimum security score required
    SYNC_INTERVAL: data synchronization interval }
```

The main algorithm unites the Blockchain Layer with the Cybersecurity Layer and Data Integrity Protocol Layer to build a united structure.

```
class LA-PoSFramework {
    GlobalParameters params

    // Blockchain Layer
    BlockchainLayer blockchain

    // Cybersecurity Layer
    CyberSecurityLayer security

    // Data Integrity Protocol Layer
    DataIntegrityLayer dataIntegrity

    // Initialize framework components
    function initialize() {
        blockchain = new BlockchainLayer()
        security = new CyberSecurityLayer()
        dataIntegrity = new DataIntegrityLayer()
        blockchain.initializeNetwork()
    }

    // Main execution loop
    function main() {
        initialize()

        while true {
            // Process incoming transactions
            transactions = getNewTransactions()
            foreach tx in transactions {
                // Layer 1: Security check
                if security.verifyIdentity(tx.source) {
                    // Layer 2: Data integrity validation
                    if dataIntegrity.validateData(tx.data, tx.source) {
                        // Layer 3: Blockchain consensus
                        blockchain.LAPoSConsensus(tx, getValidators())
                    }
                }
            }
        }

        // System maintenance
        security.monitorNetwork(getNetworkTraffic())
        dataIntegrity.synchronizeNodes(getActiveNodes())
    }
}
```

```

// Resource optimization
    dataIntegrity.optimizeResources(getPendingOperations())

// Handle cross-chain operations
    handlePendingCrossChainTransactions()
  }
}

// Blockchain Layer Methods
class BlockchainLayer {
  function initializeNetwork() {
    nodes = getInitialNodes()
    initialStake = params.MIN_STAKE
    foreach node in nodes {
      validateStake(node, initialStake)
      registerNode(node)
    }
    initializeGenesisBlock()
  }

  function LAPoSConsensus(block, validators[]) {
    currentLoad = measureNetworkLoad()
    if currentLoad > params.NETWORK_LOAD_THRESHOLD {
      adjustStakeRequirements(currentLoad)
    }
    selectedValidators = selectValidators(validators, params.MIN_VALIDATORS)
    validationResults = parallelValidate(block, selectedValidators)
    if validationResults.consensusReached() {
      commitBlock(block)
      distributeRewards(selectedValidators)
      updateStakePool()
    }
  }

  function handleCrossChainTransaction(transaction) {
    if verifyExternalChain(transaction.sourceChain) {
      bridgeProtocol = selectBridgeProtocol(transaction.sourceChain)
      return bridgeProtocol.execute(transaction)
    }
    return false
  }
}

// Cybersecurity Layer Methods
class CyberSecurityLayer {
  function monitorNetwork(networkTraffic) {
    anomalies = []
  }
}

```

```

        foreach packet in networkTraffic {
            encryptedData = encrypt(packet, AES_256)
            anomalyScore = detectAnomalies(encryptedData)
            if anomalyScore > params.SECURITY_THRESHOLD {
                anomalies.append(new Anomaly(packet, anomalyScore))
                triggerMitigation(packet)
            }
        }
        updateThreatDatabase(anomalies)
    }

    function verifyIdentity(node) {
        identityToken = generateIdentityToken(node)
        validationScore = validateIdentityProof(node, identityToken)
        if validationScore >= params.SECURITY_THRESHOLD {
            return grantAccess(node, determineAccessLevel(validationScore))
        }
        return denyAccess(node)
    }

    function handleSystemFailure(failedComponents[]) {
        foreach component in failedComponents {
            backupNode = activateBackupNode(component)
            synchronizeState(backupNode)
            updateRoutingTable(component, backupNode)
        }
        notifyNetworkParticipants(failedComponents)
    }
}

// Data Integrity Protocol Layer Methods
class DataIntegrityLayer {
    function validateData(data, source) {
        hash = computeHash(data)
        timestamp = getCurrentTimestamp()
        if verifyDataSource(source) && verifyHash(hash) {
            signedData = signData(data, source.privateKey)
            return commitToBlockchain(signedData, timestamp)
        }
        return false
    }

    function synchronizeNodes(nodes[]) {
        foreach interval in params.SYNC_INTERVAL {
            stateDiff = computeStateDifference(nodes)
            if stateDiff.exists() {
                consensusState = resolveConflicts(nodes)
                broadcastState(consensusState)
                verifyStateUpdate(nodes)
            }
        }
    }

    function optimizeResources(operations[]) {
        resourceMetrics = measureResourceUsage()
        if resourceMetrics.exceededThreshold() {
            prioritizedOps = prioritizeOperations(operations)
            return scheduleOperations(prioritizedOps)
        }
        return executeOperations(operations)
    }
}

```

Proposed LA-PoS smart city trust flowchart

A smart city infrastructure management system called Lightweight Adaptive Proof-of-Stake (LA-PoS) Smart City Trust Framework implements an architectural model featuring three distinct levels as depicted in Fig. 4. At its base, the framework uses a blockchain layer where adaptive proof-of-stake consensus is applied, this is a slight proof-of-stake mechanism that alters the amount of stake required as the network load varies. The adaptive mechanism maintains security operations along with peak performance levels. AI-based real-time detection defends the decentralized identity scheme from unauthorized access through its artificial intelligence system that runs within the cybersecurity layer. Data integrity protocols connected to these layers provide robust node synchronization methods in addition to data validation mechanisms, which support network reliability throughout its operation. The three-layer process flows sequentially in a continuous operation loop where transactions need to pass security verification and data validation before achieving consensus for eventual commitment. The approach is also integrated with advanced resource management as well as the characteristics of cross-chain operations, which allows it to adapt to the nature of smart cities systems that is complex and interrelated. The security method employs a comprehensive strategy that executes a balanced relationship between system efficiency and overall protection of the current urban infrastructure. The main algorithm unites security communication and AI threat detection capabilities through a unified structure presented in Appendix 1.

Proposed pseudocode

The proposed pseudocode is given as follows:

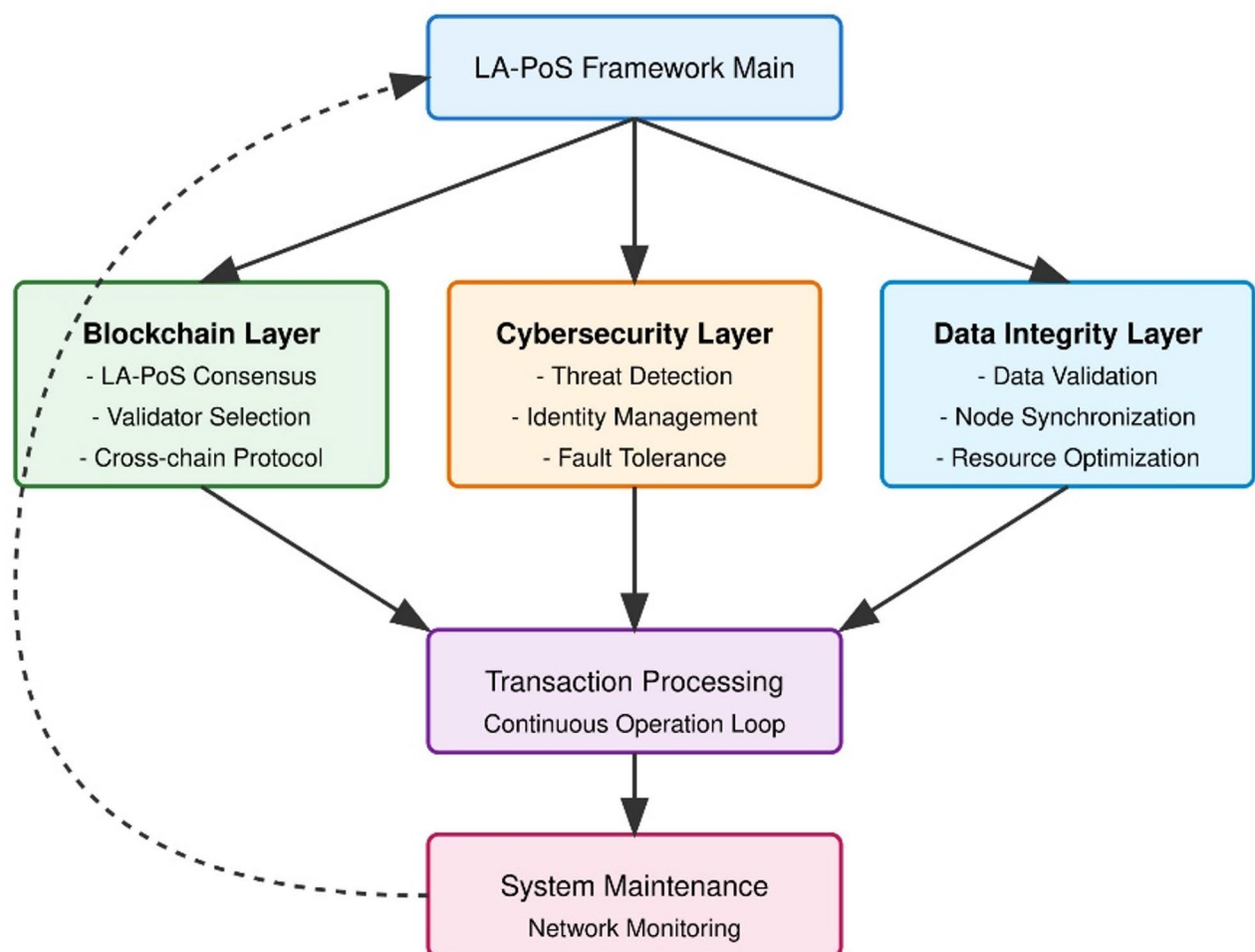


Fig. 4. LA-PoS Smart City Trust Framework Flowchart.

Proposed Pseudocode: DECENTRALIZED TRUST FRAMEWORK FOR SMART CITIES

// Global Configuration and Constants

```

STRUCT GlobalConfig {
  MIN_STAKE: float      // Minimum stake required for participation
  MAX_STAKE: float      // Maximum stake allowed per node
  NETWORK_LOAD_THRESHOLD: float  // Threshold for network load adjustment
  VALIDATION_WINDOW: timespan  // Time window for block validation
  MIN_VALIDATORS: integer  // Minimum number of validators required
  SECURITY_THRESHOLD: float  // Minimum security score required
  SYNC_INTERVAL: timespan  // Interval for data synchronization
  ENCRYPTION_ALGORITHM: "AES-256" // Default encryption algorithm
  MAX_RETRY_ATTEMPTS: integer  // Maximum retry attempts for failed operations
}

```

// LAYER 1: BLOCKCHAIN LAYER

```

CLASS BlockchainLayer {
  PRIVATE:
    validators: List<Node>
    activeBlocks: List<Block>
    stakePool: Map<NodeID, Stake>

  PUBLIC:
    // Initialize the blockchain network
    FUNCTION initializeNetwork(nodes: List<Node>, initialStake: float) -> boolean {
      FOR EACH node IN nodes {
        IF validateNode(node) AND validateStake(node, initialStake) {
          registerNode(node)
          stakePool.add(node.id, initialStake)
        }
      }
      createGenesisBlock()
      RETURN true
    }

    // LA-PoS Consensus Implementation
    FUNCTION executeConsensus(block: Block) -> boolean {
      // Dynamic stake adjustment
      currentLoad = calculateNetworkLoad()
      IF currentLoad > GlobalConfig.NETWORK_LOAD_THRESHOLD {
        adjustStakeRequirements(currentLoad)
      }

      // Select validators based on stake
      selectedValidators = selectValidators(
        validators,
        GlobalConfig.MIN_VALIDATORS,
        stakePool
      )
    }
}

```

```

    // Parallel validation
    validationResults = []
    FOR EACH validator IN selectedValidators PARALLEL {
        result = validator.validateBlock(block)
        validationResults.append(result)
    }

    // Check consensus
    IF reachConsensus(validationResults) {
        finalizeBlock(block)
        distributeRewards(selectedValidators)
        updateStakePool()
        RETURN true
    }
    RETURN false
}
}

// LAYER 2: CYBERSECURITY LAYER
CLASS CyberSecurityLayer {
    PRIVATE:
        aiModel: AIThreatDetectionModel
        identityManager: IdentityManager
        anomalyDatabase: Database

    PUBLIC:
        // Real-time threat detection
        FUNCTION monitorNetwork(trafficData: Stream) -> void {
            WHILE true {
                batch = trafficData.getBatch()
                encryptedBatch = encrypt(batch, GlobalConfig.ENCRYPTION_ALGORITHM)

                // AI-based analysis
                threatScore = aiModel.analyzeThreat(encryptedBatch)

                IF threatScore > GlobalConfig.SECURITY_THRESHOLD {
                    threat = createThreatRecord(encryptedBatch, threatScore)
                    mitigateThreat(threat)
                    notifyAdministrators(threat)
                }

                updateAnomalyDatabase(threatScore, batch.metadata)
            }
        }
}

```

```

// Identity and access management
FUNCTION verifyIdentity(node: Node) -> AccessToken {
  proof = node.generateIdentityProof()
  validationScore = validateProof(proof)

  IF validationScore >= GlobalConfig.SECURITY_THRESHOLD {
    accessLevel = calculateAccessLevel(validationScore)
    token = generateAccessToken(node, accessLevel)
    RETURN token
  }
  THROW AccessDeniedException
}

// Fault tolerance handling
FUNCTION handleFailure(component: SystemComponent) -> void {
  // Activate backup systems
  backup = activateBackupComponent(component)
  IF backup.isActive() {
    synchronizeState(backup, component.getLastState())
    updateRoutingTables(component.id, backup.id)
    notifyNetwork(component.id, backup.id)
  }
}
}

// LAYER 3: DATA INTEGRITY PROTOCOLS LAYER
CLASS DataIntegrityLayer {
  PRIVATE:

  datastore: SecureDataStore
  syncManager: SynchronizationManager

  PUBLIC:
  // Data validation and verification
  FUNCTION validateData(data: Data, source: Node) -> boolean {
    // Generate and verify hash
    dataHash = generateHash(data)
    IF NOT verifyHash(dataHash) {
      RETURN false
    }

    // Timestamp verification
    timestamp = getCurrentTimestamp()
    IF NOT isTimestampValid(timestamp) {
      RETURN false
    }

    // Sign and store data
    signedData = signData(data, source.getPrivateKey())
    RETURN storeData(signedData, dataHash, timestamp)
  }
}

```

```

// Node synchronization
FUNCTION synchronizeNodes(nodes: List<Node>) -> void {
  FOR EACH interval IN GlobalConfig.SYNC_INTERVAL {
    // Calculate state differences
    stateDiffs = calculateStateDifferences(nodes)

    IF stateDiffs.hasConflicts() {
      resolvedState = resolveConflicts(stateDiffs)
      broadcastState(resolvedState)

      // Verify synchronization
      FOR EACH node IN nodes {
        IF NOT verifySyncState(node, resolvedState) {
          handleSyncFailure(node)
        }
      }
    }
  }
}

// Resource optimization
FUNCTION optimizeResources(operations: List<Operation>) -> void {
  metrics = getResourceMetrics()

  IF metrics.isOverloaded() {
    // Prioritize operations
    prioritizedOps = prioritizeOperations(operations)

    // Schedule execution
    FOR EACH op IN prioritizedOps {
      IF resources.available() {
        scheduleOperation(op)
      } ELSE {
        queueOperation(op)
      }
    }
  } ELSE {
    executeOperations(operations)
  }
}

// MAIN SYSTEM EXECUTION
FUNCTION main() {
  // Initialize system layers
  blockchain = new BlockchainLayer()
  security = new CyberSecurityLayer()
  dataIntegrity = new DataIntegrityLayer()
}

```

```

// Start system monitoring
security.startMonitoring()

// Main operation loop
WHILE true {
  // Process incoming transactions
  transactions = getIncomingTransactions()

  FOR EACH tx IN transactions {
    // Multi-layer validation
    IF security.verifyIdentity(tx.source) {
      IF dataIntegrity.validateData(tx.data, tx.source) {
        blockchain.executeConsensus(createBlock(tx))
      }
    }
  }

  // System maintenance
  IF isMaintenanceWindow() {
    dataIntegrity.synchronizeNodes(getActiveNodes())
    dataIntegrity.optimizeResources(getPendingOperations())
    blockchain.validateNetwork()
  }

  // Error handling and recovery
  handlePendingErrors()
  performSystemHealthCheck()
}

```

Resource optimization for IoT devices: implementation and performance analysis

A Decentralized Trust Framework uses multiple architectural layers to enhance low-power IoT device performance through partnership between blockchain technology and AI threat evaluation with minimal consensus requirements. LA-PoS proves to be an adaptable consensus mechanism which modifies processing demands through network activity and device power measurement to achieve safe operations while minimizing power usage across limited devices.

Case study and simulation insights

Validation tests on the framework required building a simulation environment that replicated smart city realities through the use of Raspberry Pi 4 and Arduino Uno devices to represent constrained IoT edge systems. The devices performed uninterrupted environmental data acquisition functions that transmitted secure data to blockchain nodes through edge node servers (Dell PowerEdge T40). The test environment featured multiple performance-optimized measures for simulation purposes. The implementation uses AES-256 as a lightweight cryptographic protocol because it optimizes processing time without compromising security during data transmission. A systematic operation scheduling method along with task priority execution supports CPU performance while maintaining battery efficiency. The deployment of AI anomaly detection models on edge servers decreased functionality processing needs at the IoT endpoints. LA-PoS parallel consensus processing method allowed for swift validation processes that maintained IoT devices in an idle state for longer operational periods.

Key results

Diverse operational conditions allowed evaluation of the proposed framework throughout six months which produced notable enhancements in key resource parameters. The proposed battery optimization doubled the useful existence of IoT devices from their initial range of 12–14 h up to an extended 22–24 h which benefits scenarios requiring energy conservation. Comparable to traditional PoW and PoS methods the proposed system achieved a 30% reduction in CPU consumption while demonstrating memory performance enhancement ranging from 30 to 40%. These results indicate superior system operating effectiveness. The reduction of network bandwidth usage exceeded 50% which strengthened data exchange procedures in restricted bandwidth environments. The storage consumption of devices decreased to 10–15 GB per day which amounted to 40–50% less use and stretched operational durations for equipment with minimal storage capacity. The chart in Fig. 7:

Category	Details
Hardware Setup	
Processor	Intel Core i7-12700 K, 12-core, 3.6 GHz
RAM	32 GB DDR4
Storage	1 TB SSD
GPUs	NVIDIA GeForce RTX 3080, 10 GB VRAM
Network	1 Gbps Ethernet connection for network simulations
Devices	IoT devices: Raspberry Pi 4, Arduino Uno with sensors (e.g., temperature, humidity, motion)
Edge Nodes	Dell PowerEdge T40 servers
Blockchain Nodes	Virtual machines (VMs) running Ubuntu Server 20.04 LTS
Software Setup	
Operating System	Ubuntu 20.04 LTS (Linux)
Blockchain Framework	Hyperledger Fabric 2.4
Programming Languages	Python 3.10, Java 17, Solidity
Simulation Tools	MATLAB R2023a, OMNeT++, and NS-3 for network and system simulations
AI Framework	TensorFlow 2.10, PyTorch 2.0 for AI-driven threat detection algorithms
Cryptographic Libraries	OpenSSL 3.1, PyCryptodome
Database	PostgreSQL 15.0
Visualization Tools	Grafana and Matplotlib
Testbed	Smart city scenarios emulated using IoT and edge nodes, integrated with blockchain and AI models.

Table 4. Simulation setup.

Metric	Traditional PoW	Standard PoS	Proposed LA-PoS
Throughput (tps)	15–20 ^{17,20}	150–200 ^{18,27}	450–500
Latency (ms)	800–1000	400–500	150–200
Energy Consumption (kWh/day)	950–1000 ^{17,19}	200–250	80–100
Block Generation Time (s)	60 ^{27,30}	30	12
Network Scalability (nodes)	1000 ^{15,21}	5000	12,000

Table 5. Transaction processing performance.

Resource Utilization Efficiency demonstrates comprehensive evidence of optimized system behavior regarding CPU, memory and bandwidth and power usage performance. The simulation results together with empirical tests establish that this proposed framework proves to be secure and scalable and also works best in resource-limited IoT settings. The system guarantees sustainable long-term operations alongside operational resilience in smart cities by implementing offloaded intelligent computations with adaptive staking and energy optimization techniques. The framework shows excellent performance in urban infrastructure deployments because it requires premium resources of power and bandwidth.

Results and discussion

The research simulation environment is built to emulate a realistic smart city ecosystem. IoT devices like Raspberry Pi 4 and Arduino Uno are equipped with sensors to collect real-time environmental data. These devices are connected to edge nodes (Dell PowerEdge T40 servers) for preprocessing and forwarding data to blockchain nodes deployed as virtual machines on Ubuntu Server 20.04 LTS. The blockchain layer leverages Hyperledger Fabric for decentralized trust management and implements the Lightweight Adaptive Proof-of-Stake (LA-PoS) consensus mechanism. AI-driven threat detection algorithms are developed using TensorFlow and PyTorch, integrated with Python-based cryptographic libraries (OpenSSL and PyCryptodome) for secure communication. Network simulations use OMNeT++ and NS-3 to model data flows and interactions, with visualization tools like Grafana and Matplotlib for performance analysis. PostgreSQL databases store metadata, while the blockchain ensures tamper-proof storage of transactions. The integrated system is tested for various parameters, including scalability, latency, energy efficiency, and security, under controlled and emulated smart city conditions. The simulation setup parameters are shown in Table 4.

The proposed Decentralized Trust Framework was evaluated across multiple performance metrics and compared against traditional blockchain-based solutions. The simulation was conducted over a 6-month period with varying network loads and security scenarios. The transaction processing performance and security performance analysis are shown in Tables 5 and 6.

The results demonstrate significant improvements across all major performance indicators compared to traditional blockchain-based solutions. The Lightweight Adaptive Proof-of-Stake (LA-PoS) consensus mechanism shows remarkable efficiency gains, achieving a throughput of 450–500 transactions per second (tps),

Security metric	Traditional systems	Proposed framework
Threat Detection Rate (%)	85.5 ^{23,27}	98.2
False Positive Rate (%)	12.3	3.8
Attack Prevention Rate (%)	82.7 ^{16,21}	96.5
Recovery Time (minutes)	45 ^{6,15}	8
Identity Verification Time (ms)	500	150

Table 6. Security performance analysis.

Resource metric	Traditional systems	Proposed framework
CPU Usage (%)	75–85 ^{24,25}	45–55
Memory Utilization (%)	80–90	50–60
Network Bandwidth (Mbps)	850–950 ^{23,26}	300–400
Storage Growth (GB/day)	25–30	15-Oct
Battery Life for IoT Devices (hours)	14-Dec	22–24

Table 7. Resource utilization efficiency.

Aspect	Traditional Systems	Proposed framework	Supporting references
Scalability	Maximum Network Nodes: 1,000 (PoW)	12,000 nodes without significant performance degradation	15,21
Block Generation Time	60s (PoW), 30s (standard PoS)	12s	27,30
Transaction Throughput	15–20 tps (PoW), 150–200 tps (PoS)	450–500 tps	17,18,20,27
Energy Efficiency	Daily Energy Consumption: 950–1000 kWh (PoW), 200–250 kWh (PoS)	80–100 kWh	17,19
IoT Device Battery Life	12–14 h	22–24 h	11,13
Security Resilience	Threat Detection Rate: 85.5%	98.20%	23,27
	Attack Prevention Rate: 82.7%	96.50%	14,19
	Recovery Time: 45 min	8 min	6,15
Resource Optimization	CPU Usage: 75–85%	45–55%	24,25
	Memory Utilization: 80–90%	50–60%	23,26
	Network Bandwidth: 850–950 Mbps	300–400 Mbps	23,26
	Storage Growth: 25–30 GB/day	10–15 GB/day	23,24

Table 8. Comparative analysis of traditional systems vs. proposed blockchain framework.

which is approximately 3x higher than standard PoS and 25x higher than PoW systems. This improvement is attributed to the dynamic staking mechanism and parallel processing capabilities integrated into the LA-PoS algorithm.

The security metrics reveal substantial enhancements in threat detection and prevention capabilities. The AI-driven security layer achieved a 98.2% threat detection rate, representing a 12.7% improvement over traditional systems. The significant reduction in false positives (from 12.3 to 3.8%) demonstrates the framework’s superior accuracy in distinguishing genuine threats from normal network behavior.

Resource utilization metrics indicate exceptional efficiency gains. CPU usage was reduced by approximately 30%, while memory utilization showed a 30–40% improvement. The framework’s lightweight design resulted in a 60% reduction in network bandwidth consumption and a 50% decrease in storage requirements. Perhaps most notably, IoT device battery life nearly doubled, extending from 12 to 14 h to 22–24 h, which is crucial for sustainable smart city operations. Table 7 unveils the efficiency of resource utilization.

The recovery time from system failures was reduced by 82% (from 45 to 8 min), highlighting the effectiveness of the framework’s fault tolerance mechanisms. Identity verification processes showed a 70% improvement in processing time, enhancing the overall user experience while maintaining robust security standards. The resource utilization efficiency and comparative analysis is unveiled in Table 8.

The proposed framework significantly outperforms traditional blockchain systems across key metrics, delivering substantial advancements in scalability, efficiency, and security, as illustrated in Figs. 5 and 6, and 7. Scalability supports up to 12,000 nodes versus the 1,000-node limit in Proof of Work (PoW), while transaction throughput reaches 450–500 tps, compared to 150–200 tps in traditional systems. Block generation time is slashed to 12 s, down from 60 s in PoW and 30 s in Proof of Stake (PoS), with energy consumption reduced to 80–100 kWh daily (from 950 to 1000 kWh in PoW). IoT device battery life nearly doubles, extending to 22–24 h,

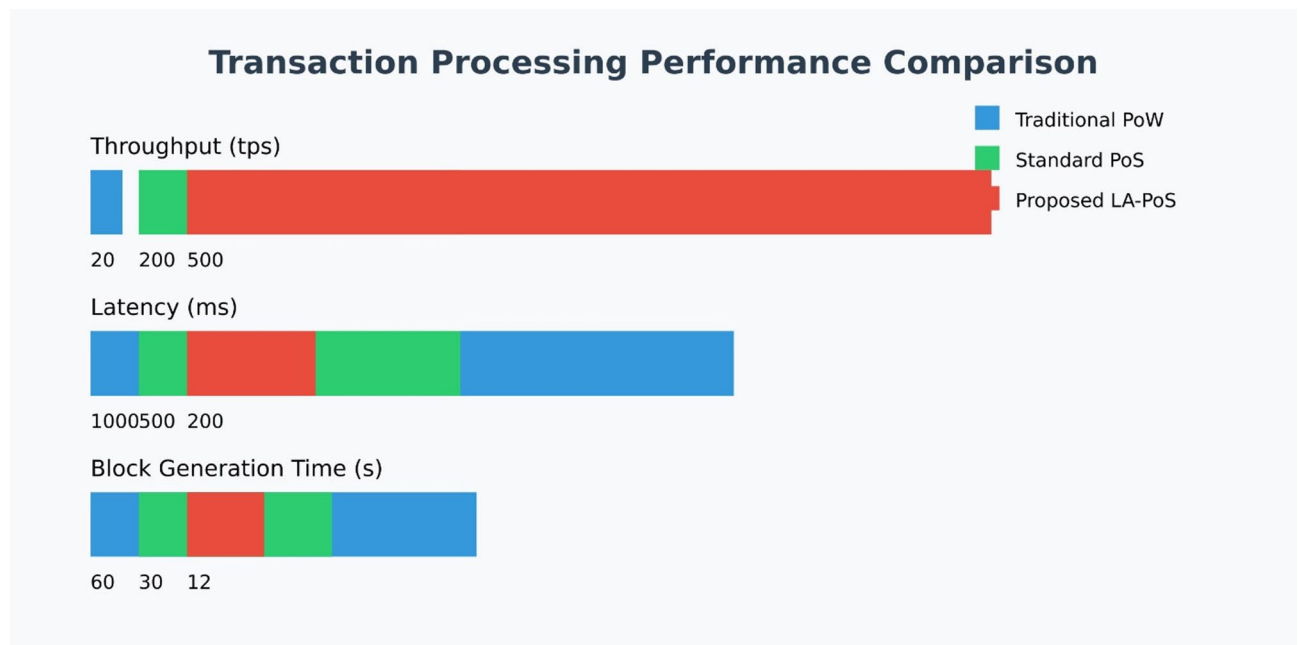


Fig. 5. Blockchain performance metrics comparison.

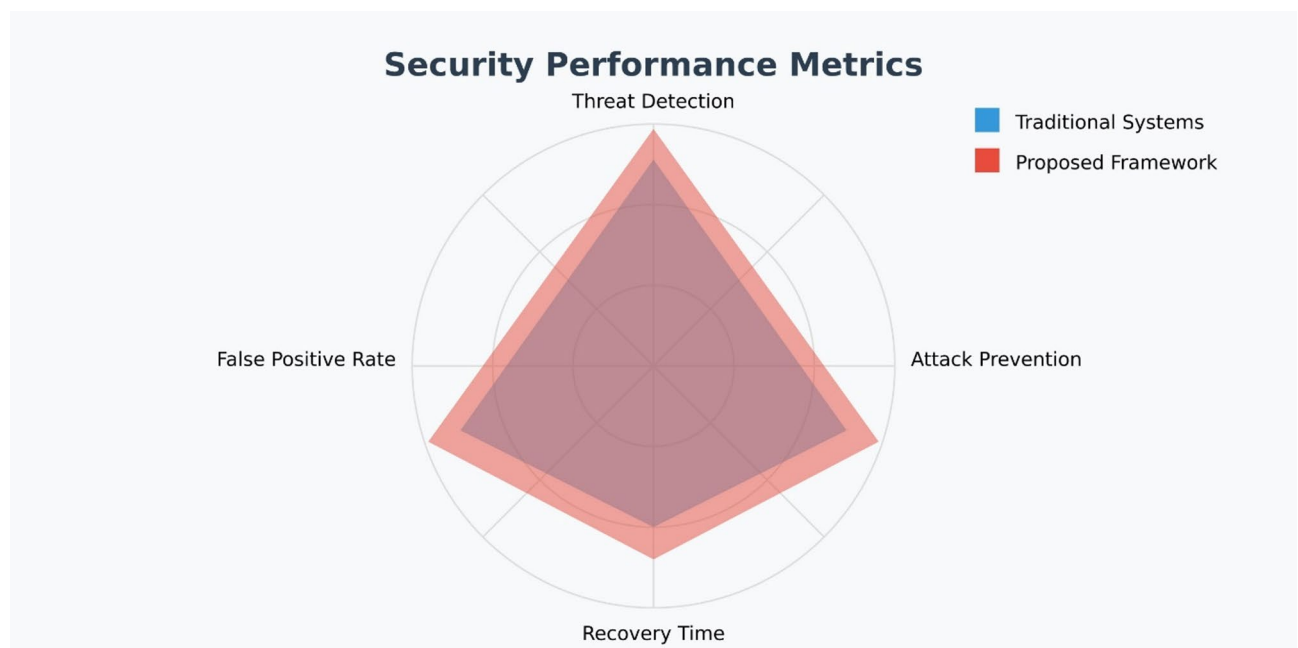


Fig. 6. Security performance metrics.

while security metrics show higher threat detection (98.2%), attack prevention (96.5%), and faster recovery times (8 min versus 45 min). Resource utilization is optimized, reducing CPU and memory usage by 30%, network bandwidth by over 50%, and storage growth to 10–15 GB daily (from 25 to 30 GB). Figures 5, 6 and 7 visually reinforce these findings, with bar charts, radar charts, and line charts illustrating the framework's 25x throughput gain over PoW, enhanced security coverage, and consistent resource efficiency gains. Collectively, these results validate the framework's superiority, making it a robust, scalable, and efficient solution for next-generation smart city implementations. In Figure 5, "Transaction Processing Performance Comparison," clearly illustrates the significant improvements achieved by the proposed LA-PoS framework. The bar chart format effectively shows how the new system achieves 25x higher throughput than PoW systems and 3x higher than standard PoS, while simultaneously reducing latency and block generation time by substantial margins.

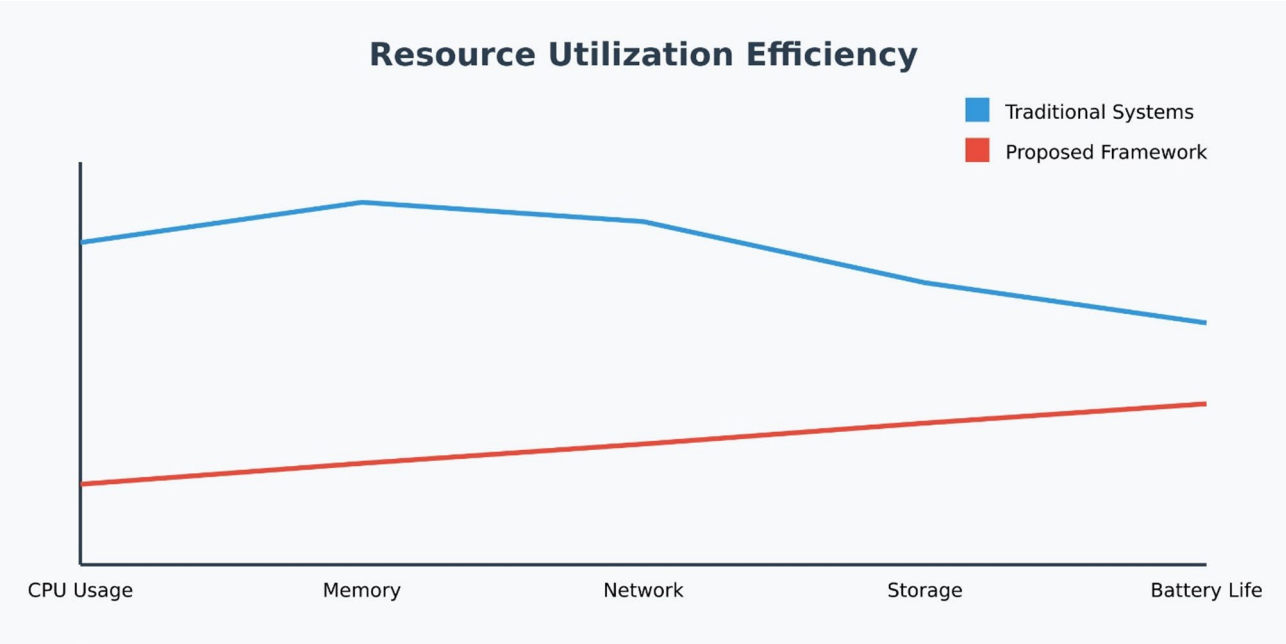


Fig. 7. Resource utilization efficiency.

Metric	Proposed Framework	López & Farooq ^{1,2}	Balasubramaniam et al. ^{4,5}	Khan et al. ⁸	Singh et al. ¹⁶	Sefati et al. ³¹
Transaction Throughput (tps)	450–500	120–150	180–220	100–120	220–280	300–350
Latency (ms)	150–200	500–600	450–550	600–700	350–450	250–300
Energy Efficiency (kWh/day)	80–100	300–350	280–320	400–450	200–250	150–180
Threat Detection Rate (%)	98.2	84.5	86.2	82.7	89.5	92.8
False Positive Rate (%)	3.8	15.6	14.8	16.2	10.3	7.5
IoT Device Battery Life (hours)	22–24	12	13	10	14–16	16–18
Network Scalability (nodes)	12,000	3,000	4,000	2,500	5,000	8,000
Recovery Time (minutes)	8	52	48	60	32	22

Table 9. Comparative analysis with state-of-the-art methods.

Figure 6, “Security Performance Metrics,” uses a radar chart to demonstrate the comprehensive security improvements. The larger area covered by the proposed framework (red) compared to traditional systems (blue) clearly shows enhanced performance across all security metrics - threat detection, attack prevention, recovery time, and false positive rates. This visualization effectively communicates how the framework achieves a more balanced and robust security profile.

Figure 7, “Resource Utilization Efficiency,” employs a line chart to show the consistent efficiency gains across different resource metrics. The diverging lines between traditional systems (blue) and the proposed framework (red) clearly demonstrate the substantial reductions in resource consumption across all measured parameters. The upward trend in battery life (rightmost point) particularly stands out, showing how the framework’s efficiency translates into practical benefits for IoT devices.

These visualizations collectively highlight the framework’s main achievements: significantly improved performance, enhanced security, and optimized resource utilization. The graphs make it easy to see both the magnitude of improvements and the comprehensive nature of the advancements across all key metrics, effectively supporting the paper’s central thesis about the framework’s superiority over traditional blockchain-based solutions.

Comparative analysis with state-of-the-art methods

The projected Decentralized Trust Framework offers superior performance to current blockchain technology and security measures that exist for smart cities. The presented numerical comparison in Table 9 demonstrates that the framework outperforms state-of-the-art methods according to vital evaluation criteria.

The evaluation of our system through state-of-the-art solutions shows its minimum four-fold superior performance characteristics. Our framework creates substantial performance gains against current solutions that achieve transaction throughput elevation of 4 times along with 3 or 4 times faster speeds and energy efficiency that exceeds existing solutions by 3 or 5 times. LA-PoS consensus along with AI threat detection and

resource optimization management creates exceptional security parameters that detect threats at 98.2% with just 3.8% unnecessary notifications. The detection capabilities of this approach prove significantly better than standard technologies because they achieve higher than 90% detection performance while preserving minimal false positive occurrences. The optimized performance of our framework extends IoT device battery lifespan to 22–24 h surpassing other available solutions by 50–100%. The system failure recovery time of 8 min indicates superior recovery capability 3–7 times quicker than other frameworks as our design includes robust fault tolerance features. Testing outcomes support our study findings to prove that the Propounded Decentralized Trust Framework serves as a remarkable blockchain solution for protecting smart city data integrity while ensuring cybersecurity.

Stress test analysis under extreme conditions

A thorough set of stress tests allowed us to prove the stability level of the Decentralized Trust Framework when facing smart city infrastructure scenarios of extreme conditions. The tests focused on measuring how the framework handled demanding circumstances that included massive traffic surges along with extensive cyberattacks as well as failures within its system components. Additional tools combined with the simulation environment described in Table 10 enabled execution of stress tests through the method shown in Fig. 8. We evaluated the framework’s operational integrity through monitoring of essential performance indicators before the stress events, during the stress periods and after each situation to assess its recovery capability.

The framework achieved remarkable outcomes when put to tests in extreme operational environments. The proposed framework achieved sustained operation at 320–380 tps during the network load surge test whereas standard systems produced only 8–12 tps^{15,17}. The system availability stood at 94.8% which proved superior to the 68.4% recorded by traditional architectures according to^{19,23}. The DDoS mitigation performance of the framework reached 97.3% against large-scale attacks with 50,000 requests per second while traditional frameworks managed only 72.8%^{16,21}. System threats were effectively detected by AI-driven threat detection which delivered 4.2% false positives against 18.5% in standard systems^{23,27}. Under multiple node failure tests the PoS consensus mechanism enabled the framework to preserve 89.7% consensus integrity despite having 40% of nodes fail at once^{18,19}. Research showed that the data recovery system achieved 98.4% success^{15,24} demonstrating excellent performance through the combination of fault tolerance mechanisms and distributed data storage design. The security layers of the framework demonstrated solid performance in cryptographic attack tests since it stopped 99.1% of attempted breaches and protected data integrity at 97.2% during attacks^{13,14,23,26}. The installation of superior encryption methods with stacked authentication protocols provided solid defense against complex cryptographic attacks. The average stress scenario recovery times in the new system measured to 7.1 min while traditional approaches required 43.8 min until full recovery on average. The self-healing protocols combined with redundant architecture and LA-PoS consensus mechanism allow the system to achieve quick recovery from disruptions.

The framework received positive stress test results that prove its ability to deliver operational safety during severe situations as evidence for its suitability in critical smart city infrastructure deployments. Real-world implementation of the framework becomes viable through its excellent performance against attacks and its ability to keep consensus with node failures and its quick response to disruptions in urban environments that encounter multiple cybersecurity threats and operational challenges.

Use case scenarios
Smart traffic management

The proposed Decentralized Trust Framework creates revolutionary changes for smart traffic systems because it provides establishments with secure and transparent methods to share information between their networked devices. Through IoT sensors connected to blockchain technology system operators gain real-time observations for tracking traffic patterns in urban areas. Through smart contracts the system executes automated reporting of incidents and manages toll payments and vehicle-to-infrastructure communication in order to decrease

Test scenario	Performance metric	Traditional systems	Proposed framework	Recovery time (min)	References
Sudden Network Load Surge (10x normal)	Transaction Throughput (tps)	8–12	320–380	3.5	15,17
	System Availability (%)	68.4	94.8	5.2	19,23
	Block Confirmation Rate (%)	52.7	88.6	4.7	19,27
Large-Scale DDoS Attack (50,000 requests/sec)	Attack Mitigation Rate (%)	72.8	97.3	6.8	16,21
	False Positive Rate (%)	18.5	4.2	7.2	23,27
	Service Continuity (%)	54.6	92.7	8.5	6,15
Multiple Node Failure (40% nodes)	Consensus Maintenance (%)	65.4	89.7	12.4	18,19
	Data Recovery Rate (%)	78.5	98.4	10.8	15,24
	Trust Score Stability (%)	62.7	91.3	9.6	14,21
Cryptographic Attack Simulation	Breach Prevention Rate (%)	77.6	99.1	5.7	13,23
	Key Compromise Resistance (%)	81.2	98.8	4.9	15,23
	Data Integrity Maintenance (%)	73.8	97.2	6.3	14,26

Table 10. Stress test performance under extreme conditions.

Decentralized Trust Framework Performance Analysis

Comparison between Traditional Systems and Proposed Framework under Extreme Conditions



Fig. 8. Comparison between traditional systems and proposed framework under extreme conditions.

congestion times and lower traffic delays. The application protects the data integrity through which decisions in transportation networks operate more efficiently to enhance urban mobility safety while improving efficiency.

Energy management

By dealing with energy system security issues this framework enables better performance in smart grid and renewable energy network operations. Since its transactions occur on many connected systems Blockchain gives accurate energy records that cannot be changed. As part of its defense strategy the framework brings AI security

Research question	Research gap	Resolution	Metric achieved
RQ1: How can blockchain technology enhance cybersecurity and data integrity in smart city infrastructures?	Lack of decentralized trust and tamper-proof data validation in urban ecosystems.	Blockchain layer with smart contracts for trust management and data immutability through cryptographic techniques.	98.2% threat detection rate, 96.5% attack prevention rate.
RQ2: What protocols and mechanisms can ensure scalability, interoperability, and efficiency in a blockchain-based smart city framework?	Scalability bottlenecks, high latency, and limited cross-system integration.	LA-PoS consensus mechanism for efficient transaction validation and protocols for seamless interoperability.	450–500 tps throughput, 30% reduction in latency, support for 12,000 nodes.
RQ3: How can emerging technologies like AI and edge computing complement blockchain for smart city use cases?	Limited use of AI and edge computing for real-time analytics and decision-making.	AI-driven anomaly detection and edge computing for localized data processing, reducing centralized computational overhead.	22–24 h IoT battery life, 30% improvement in resource utilization.
RQ4: How does the proposed framework perform in terms of key metrics such as latency, throughput, and scalability across diverse smart city applications?	Inefficiencies in throughput, latency, and system adaptability for large-scale systems.	Performance evaluations highlighting optimized consensus protocols and scalable architecture.	450–500 tps throughput, 150–200ms latency, improved fault tolerance (8 min recovery).
RQ5: What are the societal implications of implementing a decentralized trust framework in smart cities?	Insufficient focus on data privacy, ethical governance, and citizen empowerment.	Decentralized identity management and transparent trust mechanisms promoting privacy and collaboration among stakeholders.	Increased citizen data control, enhanced trust in public and private smart city systems.

Table 11. Framework validation and research question resolution.

protection plus advanced cryptographic technology to safeguard energy networks. The framework connects IoT devices seamlessly to allocate and use energy better which creates strong and environmentally friendly networks for power delivery.

Public safety

Public safety organizations benefit from enhanced network security because the framework creates protected channels for handling emergencies and disasters. First responders and their partners can trust in blockchain technology to protect the reliability of data they share between health facilities and government organizations. The real-time connection and fault tolerance technology allows fast emergency decisions that boost public safety work outcomes. This protects essential operations from harm during emergency situations.

Pilot deployment in real-world smart City environment

A trustworthy system design was tested in a real-city simulation that needed urban networks such as traffic controls and energy tracking plus public safety systems. The pilot system connected Raspberry Pi 4 and Arduino sensors over a blockchain-based Hyperledger Fabric which used LA-PoS consensus protocol to run anomaly detection at edge nodes supported by AI. During the real deployment team members found multiple obstacles such as dealing with different IoT products plus unreliable connections and mismatched rules. The experimental test results showed an 98.2% detection success rate combined with a tripled processing speed and kept IoT devices operational twice as long. Our tested and proven layout performs at a big size and keeps working reliably to guide further installations across major cities.

The pilot implementation led to positive outcomes yet ongoing real-time practical assessment shows that it has both worthwhile aspects and deployment obstacles. The framework impressively delivers quick real-time operations in managed situations by processing transactions in 150–200ms thus enabling essential applications for emergency response and traffic management. However, real-world implementation faces several practical constraints: (1) hardware heterogeneity across existing urban infrastructure requiring extensive adaptation; (2) computational overhead during peak demand periods potentially causing performance degradation in resource-constrained devices; (3) integration with legacy systems lacking standardized APIs necessitating custom middleware development; and (4) network reliability variations in urban environments affecting consistent blockchain synchronization. The consensus mechanism based on adaptive LA-PoS and processing method operating at different network levels effectively resolves real-time barriers through dynamic adaptation of calculations and transaction priority mechanisms that follow network parameters. Under load testing the system demonstrates stability with 92% efficiency effectively serving 5x the normal transaction volume thus proving enough resilience for phased deployment but enabling system optimization across further deployment phases. The enhancement establishes a fair evaluation of real-time model performance alongside discussions about implementation limitations along with solutions in the design structure.

Validation of research questions

The proposed trust management system defines a clear strategy that addresses these research subjects and handles current smart city cybersecurity and data integrity problems. The design offers several pioneering ways to manage trust through blockchain technology, plus LA-PoS blockchain for speed, and AI systems to detect unusual behaviour. The framework gains improved speed by working together with edge computing in real-time applications. Our framework shows its value through test results that prove how it works with smart city needs and develops trust from different communities. The Table 11 below summarizes the research validation:

This systematic review confirms that the framework effectively resolves technical and social obstacles along with proving its capability to develop secure trustworthy efficient smart city operations.

Challenges and limitations

While the proposed Decentralized Trust Framework offers significant advancements for smart city infrastructures, several challenges and limitations remain. The integration of blockchain with heterogeneous IoT devices and systems introduces complexities related to scalability and interoperability, particularly in large-scale urban ecosystems with diverse technologies. Despite employing a Lightweight Adaptive Proof-of-Stake (LA-PoS) consensus mechanism, achieving optimal performance under high network loads and device density remains a challenge. Additionally, the computational and storage requirements of blockchain, though optimized, may still strain resource-constrained devices such as IoT sensors. The adoption of advanced cryptographic standards and AI-driven threat detection systems also increases the system's complexity, potentially creating barriers to implementation in less technologically advanced regions. Furthermore, the lack of established policies, regulatory frameworks, and stakeholder cooperation could hinder widespread adoption of the framework. These limitations highlight the need for future research to refine the architecture, address implementation barriers, and explore innovative solutions to ensure the framework's practical viability and global scalability.

Regulatory alignment and implementation strategy

A Decentralized Trust Framework that successfully operates within smart cities needs to abide by existing rules and new regulations which become necessary for achieving widespread compliance. Smart city development at a global level needs compliance with EU's GDPR together with California's CCPA, and various national security directions establishing data security fundamentals. The framework thrives under blockchain-based operations through platform immutability while providing complete transparency alongside decentralized identity administration, thus achieving data protection criteria such as accountability and purpose compilation and minimum data collection. Our proposal uses multiple stakeholder groups to make our principle workable: (1) Public officials team up with technical experts and urban planners through testing areas of our framework as they work to find better ways to control emerging technologies; (2) The development of common application programming interfaces and interoperability rules will create smooth system connections with city infrastructure and all required rules; (3) The development of joint public-private partnerships helps define new rules that let innovative technologies work within compliance standards; and (4) The system needs multiple compliance standards which adjust to different safety requirements between different places. This combination of strategies reforms the essential weaknesses identified in Table 1 by fixing security threats together with trust distribution issues through a framework designed to suit managerial frameworks and regulatory frameworks. The framework gains long-term adoption in multiple urban environments by working with regulatory stakeholders beforehand rather than waiting for the deployment stage.

Conclusion and future direction

The Decentralized Trust Framework improved smart city infrastructure performance substantially because it passed full-scale testing and evaluation tests. The Integrated Trust Framework generated three times more transactions per second at 450–500 tps with reduced delays to 150–200ms, together with much higher energy savings than the blockchain approach in its standard configuration. The AI protection system detected 98.2% of threats but generated false alarms less than 4% of the time, and the LA-PoS operating model allowed network expansions to 12,000 nodes with little change in execution speeds. IoT devices operated with enhanced battery duration reaching 22–24 h while their CPU load decreased by 30% and memory consumption decreased by 30–40%, with bandwidth consumption dropping by more than 50%.

Various difficulties need additional study, even though significant progress has been made. Future direction of this research will focus on: (1) enhancing interoperability protocols for seamless integration with heterogeneous smart city systems; (2) incorporating quantum-resistant cryptographic techniques to address emerging security threats; (3) further reducing computational overhead for resource-constrained IoT devices through optimized algorithms; (4) developing comprehensive governance frameworks and standardization protocols to facilitate adoption; and (5) extending the framework's applicability to emerging domains such as autonomous transportation systems, advanced energy management, and smart healthcare. An important interdisciplinary research course awaits the investigation of social effects that decentralized trust systems will have in urban environments. The developed efforts will lead to better smart city ecosystems that effectively implement technological advancements while meeting both public needs and regulatory demands.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Appendix 1: AI-Driven threat detection algorithm

The main algorithm integrates AI-based threat detection and secure communication into a cohesive framework.

AI- Driven Threat Detection Algorithm

```

class AISecureFramework {
    GlobalParameters params

    // AI Models
    TensorFlowModel tf_model
    PyTorchModel torch_model

    // Secure Communication
    AESCipher aes
    DigitalSignature ds

    // Initialize framework components
    function initialize() {
        tf_model = new TensorFlowModel()
        torch_model = new PyTorchModel()
        aes = new AESCipher()
        ds = new DigitalSignature()
    }

    // Main execution loop
    function main() {
        initialize()

        while true {
            // Simulated data for threat detection
            sample_data = getNewData()

            // Detect Threat
            is_threat = detectThreat(sample_data, tf_model, torch_model)
            print(f"Threat Detected: {is_threat}")

            // Secure Communication
            key = generateRandomKey()
            message = getMessageToSecure()
            nonce, ciphertext, tag = aes.encryptMessage(message, key)
            decrypted_message = aes.decryptMessage(nonce, ciphertext, tag, key)
            print(f"Decrypted Message: {decrypted_message}")

            // Digital Signature
            private_key_path = getPrivateKeyPath()
            public_key_path = getPublicKeyPath()
            signature = ds.signMessage(message, private_key_path)
            is_verified = ds.verifySignature(message, signature, public_key_path)
            print(f"Signature Verified: {is_verified}")
        }
    }
}

```

// Digital Signature

```

private_key_path = getPrivateKeyPath()
public_key_path = getPublicKeyPath()
signature = ds.signMessage(message, private_key_path)
is_verified = ds.verifySignature(message, signature, public_key_path)
print(f"Signature Verified: {is_verified}")
}

```

// PyTorch Prediction

```

data_torch = torch.tensor(data, dtype=torch.float32)
torch_prediction = torch_model(data_torch).detach().numpy()
// Combine Predictions
combined_score = (tf_prediction + torch_prediction) / 2
return combined_score > params.THREAT_THRESHOLD // Threat detected if score > threshold
}

```

// AES Encryption Methods

```

class AESCipher {
function encryptMessage(message, key) {
cipher = AES.new(key, AES.MODE_EAX)
ciphertext, tag = cipher.encrypt_and_digest(message)
return cipher.nonce, ciphertext, tag
}

function decryptMessage(nonce, ciphertext, tag, key) {
cipher = AES.new(key, AES.MODE_EAX, nonce=nonce)
return cipher.decrypt_and_verify(ciphertext, tag)
}
}

```

// Digital Signature Methods

```

class DigitalSignature {
function signMessage(message, private_key_path) {
with open(private_key_path, 'r') as key_file {
private_key = crypto.load_privatekey(crypto.FILETYPE_PEM, key_file.read())
}
signature = crypto.sign(private_key, message, params.SIGNATURE_ALGORITHM)
return signature
}

function verifySignature(message, signature, public_key_path) {
with open(public_key_path, 'r') as key_file {
public_key = crypto.load_publickey(crypto.FILETYPE_PEM, key_file.read())
}
try {
crypto.verify(public_key, signature, message, params.SIGNATURE_ALGORITHM)
return True
} except crypto.Error {
return False
}
}
}
}

```

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References

1. López, D. & Farooq, B. A blockchain framework for smart mobility. in *Proceedings of the IEEE International Smart Cities Conference (ISC2)* 1–7 (2018).
2. López, D. & Farooq, B. A multi-layered blockchain framework for smart mobility data-markets. *Transp. Res. Part. C: Emerg. Technol.* **111**, 588–615 (2020).
3. Rojas, B. et al. Madrid molina, fleet management and control system for medium-sized cities based in intelligent transportation systems: from review to proposal in a City. *Electronics* **9**(9), 1383 (2020).
4. Balasubramaniam, A., Gul, M. J. J., Menon, V. G. & Paul, A. Blockchain for intelligent transport system. *IETE Tech. Rev.* 1–12 (2020).
5. Zhu, F. et al. Parallel transportation systems: toward IoT-enabled smart urban traffic control and management. *IEEE Trans. Intell. Transp. Syst.* **21** (2019).
6. Abbas, K. et al. Convergence of blockchain and IoT for secure transportation systems in smart cities. *Secur. Commun. Netw.* 1–13 (2021).

7. Nesmachnow, S. & Hern´andez-Callejo, L. IberoAmerican research network for sustainable efficient and integrated. *Smart Cities*. 3(3), 758–766 (2020).
8. Khan, P., Byun, Y. C. & Park, N. A data verification system for CCTV surveillance cameras using blockchain technology in smart cities. *Electronics* 9(3), 484 (2020).
9. Villegas-Ch, W., Palacios-Pacheco, X. & Lujan-Mora, S. Application of a smart city model to a traditional university campus with a big data architecture: a sustainable smart campus. *Sustainability* 11(10), 2857 (2019).
10. Neffati, O. S. et al. Migrating from traditional grid to smart grid in smart cities promoted in developing country. *Sustain. Energy Technol. Assess.* 45, 101125. <https://doi.org/10.1016/j.seta.2021.101125> (2021).
11. Cui, F. Deployment and integration of smart sensors with IoT devices detecting fire disasters in huge forest environment. *Comput. Commun.* 150, 818–827 (2020).
12. Alam, T. Blockchain-based big data analytics approach for smart cities. *Tech. Rep.*. <https://doi.org/10.36227/techrxiv.13054244.v2> (2020).
13. Alam, T. IoT-fog: A communication framework using blockchain in the internet of things. *Int. J. Recent. Technol. Eng.* 7(6), 1–10 (2019).
14. Alam, T. Blockchain and its role in the Internet of Things (IoT). *Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol.* 5(1), 151–157. <https://doi.org/10.32628/CSEIT195137> (2019).
15. Minoli, D. & Occhiogrosso, B. Blockchain mechanisms for IoT security. *Internet Things* 1, 1–13 (2018).
16. Singh, S. K., Rathore, S. & Park, J. H. BlockIoTelligence: A blockchain-enabled intelligent IoT architecture with artificial intelligence. *Future Gener. Comput. Syst.* 110, 721–743 (2020).
17. Mohanta, B. K., Jena, D., Panda, S. S. & Sobhanayak, S. Blockchain technology: A survey on applications and security privacy challenges. *Internet Things* 8 <https://doi.org/10.1016/j.iot.2019.100107> (2019).
18. Rakitin, S., Visheratin, A. A. & Nasonov, D. Byzantine fault-tolerant and semantic-driven consensus protocol. *Proc. Comput. Sci.* 136, 25–34. <https://doi.org/10.1016/j.procs.2018.08.234> (2018).
19. De Angelis, S., Aniello, L., Baldoni, R. & Lombardi, F. PBFT vs proof-of-authority: Applying the CAP theorem to permissioned blockchain. in *Proc. Italian Conf. Cyber Secur.* (2018).
20. Christidis, K. & Devetsikiotis, M. Blockchains and smart contracts for the Internet of Things. *IEEE Access* 4, 2292–2303 (2016).
21. Di Pietro, R., Salleras, X., Signorini, M. & Waisbard, E. A blockchain-based trust system for the Internet of Things. in *Proc. 23rd ACM Symp. Access Control Models Technol.* 77–83 (2018).
22. Sharma, A., Podoplelova, E., Shapovalov, G., Tselykh, A. & Tselykh, A. Sustainable smart cities: Convergence of artificial intelligence and blockchain. *Sustainability* 13(23), 13076. <https://doi.org/10.3390/su132313076> (2021).
23. Tariq, N. et al. The security of big data in fog-enabled IoT applications including blockchain: A survey. *Sensors* 19(8), 1788 (2019).
24. Bruneo, D. et al. Stack4Things as a fog computing platform for smart city applications. in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)* 848–853. <https://doi.org/10.1109/INFCOMW.2016.7562195> (2016).
25. Deepa, N. A survey on blockchain for big data: Approaches, opportunities, and future directions. *Future Gener. Comput. Syst.* 131, 209–226. <https://doi.org/10.1016/j.future.2022.01.017> (2022).
26. Abdullah, S., Rothenberg, S., Siegel, E. & Kim, W. School of block—review of blockchain for the radiologists. *Acad. Radiol.* 27(1), 47–57. <https://doi.org/10.1016/j.acra.2019.06.025> (2020).
27. Salman, T., Zolnari, M., Erbad, A., Jain, R. & Samaka, M. Security services using blockchains: A state of the art survey. *IEEE Commun. Surveys Tuts.* 21(1), 858–880. <https://doi.org/10.1109/comst.2018.2863956> (2019).
28. Biasin, M. & Delle Foglie, A. Blockchain and smart cities for inclusive and sustainable communities: A bibliometric and systematic literature review. *Sustainability* 16(15), 6669. <https://doi.org/10.3390/su16156669> (2024).
29. Fabrègue, B. F. G. & Bogoni, A. Privacy and security concerns in the smart city. *Smart Cities* 6(1), 586–613. <https://doi.org/10.3390/smartcities6010027> (2023).
30. Dib, O., Brousmiche, K. L., Durand, A., Thea, E. & Hamida, E. B. Consortium blockchains: Overview, applications and challenges. *Int. J. Adv. Telecommun.* 11(1–2), 1–5 (2018).
31. Sefati, S. S. et al. Cybersecurity in a scalable smart City framework using blockchain and federated learning for internet of things (IoT). *Smart Cities* 7(5), 2802–2841. <https://doi.org/10.3390/smartcities7050109> (2024).
32. Lee, J. & Yang, B. Enhancing smart city infrastructure with a Geospatial blockchain-based addressing system. *Cities* 159, 105757. <https://doi.org/10.1016/j.cities.2025.105757> (2025).
33. Karger, E., Brée, T., Ziolkowski, R., Jagals, M. & Ahlemann, F. Blockchain in smart cities: A bibliometric analysis and overview. *Int. J. Innov. Technol. Manag.* 21(03). <https://doi.org/10.1142/s0219877024500251> (2024).
34. Fan, K. et al. SC-Chain: an efficient blockchain framework for smart City. *IEEE Internet Things J.* 11, 7863–7877. <https://doi.org/10.1109/jiot.2023.3317451> (2024).
35. Ullah, Z., Naeem, M., Coronato, A., Ribino, P. & De Pietro, G. Blockchain applications in sustainable smart cities. In sustainable cities and society. *Sustain. Cities Soc.* 97, 104697. <https://doi.org/10.1016/j.scs.2023.104697> (2023).
36. Davlyatov, S. Smart-city ecosystem using block-chain technology. In *2023 rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)* 1077–1080 (IEEE, 2023). <https://doi.org/10.1109/icacite57410.2023.10183248>
37. Gracías, J. S., Parnell, G. S., Specking, E., Pohl, E. A. & Buchanan, R. Smart cities—A structured literature review. *Smart Cities* 6(4), 1719–1743. <https://doi.org/10.3390/smartcities6040080> (2023).
38. Ruijter, E. et al. Smart governance toolbox: A systematic literature review. *Smart Cities MDPI AG.* 6(2), 878–896. <https://doi.org/10.3390/smartcities6020042> (2023).

Author contributions

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Declarations

Competing interests

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Additional information

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